

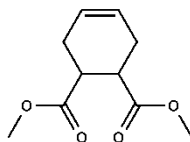
- 1 A sample of element X consists of only three isotopes:  $^{68}\text{X}$ ,  $^{70}\text{X}$ , and  $^{72}\text{X}$ . The relative atomic mass ( $A_r$ ) of X in this sample is 70.2.

| isotope         | percentage abundance / % |
|-----------------|--------------------------|
| $^{68}\text{X}$ | $x$                      |
| $^{70}\text{X}$ | 30                       |
| $^{72}\text{X}$ | $y$                      |

What is the value of  $y$ ?

- A 20  
B 30  
C 35  
D 40  
E 45
- 2 Magnesium is an element in Group 2 of the Periodic Table.  
Which of the following statements is/are correct about magnesium?
- 1 Magnesium is a stronger reducing agent than calcium.  
2 Magnesium has a smaller atomic radius than calcium.  
3 Magnesium reacts with nitrogen gas to form a compound containing 72% magnesium by mass.  
( $A_r$  values: N = 14; Mg = 24; Ca = 40)
- A none of them  
B 1 only  
C 2 only  
D 3 only  
E 1 and 2 only  
F 1 and 3 only  
G 2 and 3 only  
H 1, 2 and 3

- 3 The skeletal formula of an organic compound is shown below.  
What is the empirical formula of this compound?



- A  $\text{C}_5\text{H}_7\text{O}_2$   
 B  $\text{C}_{10}\text{H}_{14}\text{O}_4$   
 C  $\text{C}_5\text{H}_6\text{O}_2$   
 D  $\text{C}_5\text{H}_8\text{O}_2$   
 E  $\text{C}_4\text{H}_7\text{O}_2$   
 F  $\text{C}_2\text{H}_3\text{O}$
- 4 After water vapour and carbon dioxide have been removed, liquefied air contains mainly nitrogen, argon and oxygen. Their boiling points are shown.

As the liquid is warmed slowly, in which order are the gases collected first to last?

| gas      | boiling point / $^{\circ}\text{C}$ |
|----------|------------------------------------|
| nitrogen | $-196^{\circ}\text{C}$             |
| argon    | $-186^{\circ}\text{C}$             |
| oxygen   | $-183^{\circ}\text{C}$             |

- A oxygen, argon, nitrogen  
 B nitrogen, oxygen, argon  
 C nitrogen, argon, oxygen  
 D argon, nitrogen, oxygen  
 E argon, oxygen, nitrogen

- 5** Two experiments were carried out using separate samples of an aqueous solution of an unknown base B of concentration  $0.150 \text{ mol dm}^{-3}$  ( $11.1 \text{ g dm}^{-3}$ ), and the following observations were made:

\*  $20.0 \text{ cm}^3$  of the base solution was completely neutralised by  $30.0 \text{ cm}^3$  of  $0.200 \text{ mol dm}^{-3}$  hydrochloric acid,  $\text{HCl(aq)}$ .

\* A pale blue precipitate formed when a few drops of aqueous copper(II) sulfate,  $\text{CuSO}_4\text{(aq)}$ , were added to a portion of the base solution.

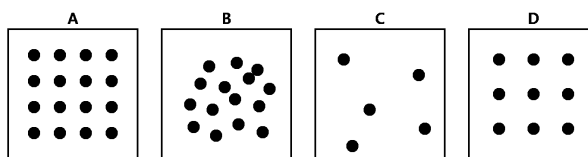
Which of the following statements can be deduced from this information?

1. One mole of the base reacts with two moles of  $\text{H}^+$  ions in complete neutralisation.
2. The base is fully dissociated in aqueous solution.
3. The relative formula mass ( $M_r$ ) of the base is 74.

( $A_r$  values are not required)

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3
- 
- 6** What are the oxidation states of sulfur in  $\text{SO}_2$  and  $\text{SO}_4^{2-}$ , respectively?
- A** +2 and +4
- B** +4 and +6
- C** +4 and +4
- D** +6 and +4
- E** -4 and -6
- F** 0 and +6

- 7 Which particle diagram best represents a liquid? The particles are shown at one instant and are not drawn to scale.



- A** A  
**B** B  
**C** C  
**D** D
- 8 Pollutant P is formed when a carbon-containing fuel burns with insufficient oxygen. Pollutant Q can form inside a hot car engine when nitrogen and oxygen from the air react.

Which pair correctly identifies P and Q?

- A** P: carbon monoxide; Q: nitrogen oxides  
**B** P: carbon dioxide; Q: sulfur dioxide  
**C** P: sulfur dioxide; Q: carbon monoxide  
**D** P: nitrogen oxides; Q: carbon dioxide  
**E** P: methane; Q: carbon monoxide
- 9 A diol reacts with an excess of propanoic acid in the presence of an acid catalyst to form a diester.

The diester has a relative molecular mass ( $M_r$ ) of 174.

What is the relative molecular mass ( $M_r$ ) of the diol?

( $A_r$  values: H = 1.0; C = 12.0; O = 16.0)

- A** 26  
**B** 44  
**C** 62  
**D** 76  
**E** 90  
**F** 100  
**G** 118  
**H** 136

- 10** Researchers intend to isolate mitochondria from a liver cell homogenate using differential centrifugation. The table below displays the minimum gravitational force (g) required to pellet specific cellular components.

| Component    | Minimum Force to Pellet (g) |
|--------------|-----------------------------|
| Nuclei       | 1,000                       |
| Mitochondria | 15,000                      |
| Ribosomes    | 100,000                     |

The homogenate is first centrifuged at 1,000 g for 10 minutes, producing a pellet and a supernatant.

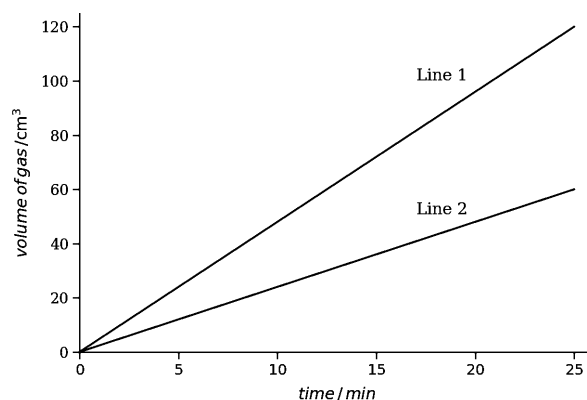
Which of the following subsequent steps would yield the purest sample of mitochondria?

- A** Resuspend the pellet from the first step and centrifuge at 15 000 *g*.
  - B** Centrifuge the supernatant from the first step at 15 000 *g* and collect the resulting pellet.
  - C** Centrifuge the supernatant from the first step at 15 000 *g* and collect the resulting supernatant.
  - D** Centrifuge the supernatant from the first step at 100 000 *g* and collect the resulting pellet.
  - E** Centrifuge the supernatant from the first step at 100 000 *g* and collect the resulting supernatant.
- 11** An aqueous solution gives a lilac flame. After acidification with dilute nitric acid, aqueous silver nitrate gives a cream precipitate. Which compound could be present?
- A** LiCl
  - B** NaBr
  - C** KCl
  - D** KBr
  - E**  $\text{CaI}_2$
  - F**  $\text{CuBr}_2$

- 12 A pure metal and an alloy made from it have the properties shown. Which statement is best supported by the data?

| material   | tensile strength / MPa | percentage extension before breaking |
|------------|------------------------|--------------------------------------|
| pure metal | 210                    | 42                                   |
| alloy      | 510                    | 13                                   |

- A The alloy is weaker and more ductile than the pure metal.  
 B The alloy is stronger and less ductile than the pure metal.  
 C The alloy must conduct electricity better than the pure metal.  
 D The alloy must have a lower density than the pure metal.  
 E The two materials have identical mechanical properties.
- 13 An aqueous solution of dilute sulfuric acid,  $\text{H}_2\text{SO}_4(\text{aq})$ , is electrolysed at room temperature and pressure using inert platinum electrodes and a constant electric current.
- The graph shows the volume of gas collected at each of the two electrodes as a function of time.
- Which of the following statements is/are correct?
- Line 1 represents the gas collected at the cathode.
  - In the external circuit, electrons flow from the electrode producing the gas of Line 1 towards the electrode producing the gas of Line 2.
  - Doubling the electric current would double the gradient of both Line 1 and Line 2.



- A none of them  
 B 1 only  
 C 2 only  
 D 3 only  
 E 1 and 2 only  
 F 1 and 3 only  
 G 2 and 3 only  
 H 1, 2 and 3

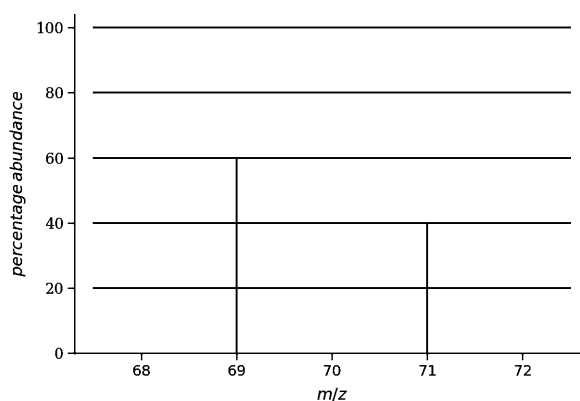
- 14 The mass spectrum of a sample of element M is shown in the diagram.

A 0.100 mol sample of M reacts completely with 2.40 g of oxygen gas ( $\text{O}_2$ ) to form a single oxide.

Which row in the table gives the relative atomic mass of this sample of M and the empirical formula of the oxide formed?

( $A_r(\text{O}) = 16.0$ )

|   | Relative atomic mass | Empirical formula of oxide |
|---|----------------------|----------------------------|
| A | 69.8                 | MO                         |
| B | 69.8                 | $\text{M}_2\text{O}_3$     |
| C | 69.8                 | $\text{MO}_2$              |
| D | 70.0                 | $\text{M}_2\text{O}_3$     |
| E | 70.0                 | $\text{MO}_2$              |
| F | 70.2                 | MO                         |
| G | 70.2                 | $\text{M}_2\text{O}_3$     |
| H | 70.2                 | $\text{MO}_2$              |



**15** Which of the following statements is/are correct?

1. Ethene decolourises bromine water.
2. Addition polymerisation of ethene produces a small-molecule by-product.
3. Ethanol reacts with sodium to release hydrogen gas.

- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

**16** A solution contains both hydrochloric acid,  $\text{HCl(aq)}$ , and sulfuric acid,  $\text{H}_2\text{SO}_4\text{(aq)}$ .

- A  $50.0\text{ cm}^3$  sample of this solution is completely neutralized by  $35.0\text{ cm}^3$  of  $0.200\text{ mol dm}^{-3}\text{ NaOH(aq)}$ .

- A separate  $50.0\text{ cm}^3$  sample of the same solution is treated with excess aqueous barium chloride,  $\text{BaCl}_2\text{(aq)}$ , producing  $0.466\text{ g}$  of dry  $\text{BaSO}_4\text{(s)}$  precipitate.

What is the concentration of hydrochloric acid in the original solution?

( $A_r$  values:  $\text{H} = 1.0$ ;  $\text{O} = 16.0$ ;  $\text{S} = 32.0$ ;  $\text{Ba} = 137.0$ )

- A**  $0.020\text{ mol dm}^{-3}$   
**B**  $0.030\text{ mol dm}^{-3}$   
**C**  $0.040\text{ mol dm}^{-3}$   
**D**  $0.060\text{ mol dm}^{-3}$   
**E**  $0.080\text{ mol dm}^{-3}$   
**F**  $0.100\text{ mol dm}^{-3}$

- 17 Three separate experiments are carried out. In each experiment, an excess of dilute hydrochloric acid is reacted with 0.050 mol of a solid Group 1 carbonate:

- Experiment 1:  $\text{Li}_2\text{CO}_3$
- Experiment 2:  $\text{Na}_2\text{CO}_3$
- Experiment 3:  $\text{K}_2\text{CO}_3$

Let  $V_1$ ,  $V_2$ , and  $V_3$  be the final volumes of gas produced in Experiments 1, 2, and 3 respectively, measured at room temperature and pressure.

Which of the following correctly compares these volumes?

- A  $V_1 < V_2 < V_3$
- B  $V_1 > V_2 > V_3$
- C  $V_1 = V_2 = V_3$
- D  $V_1 = V_2 < V_3$
- E  $V_1 > V_2 = V_3$

- 18 The ions  $\text{X}^{2-}$ ,  $\text{Y}^-$  and  $\text{Z}^{2+}$  are isoelectronic (they have the same electronic configuration).

Which of the following statements about the **neutral atoms** X, Y and Z is correct?

- A Element Z has the lowest atomic number.
- B Element Z is in the period immediately following elements X and Y.
- C Elements X, Y and Z are in the same period.
- D Element X is in Group 2 of the Periodic Table.
- E Element Y has a larger atomic radius than element Z.

- 19 Electrolysis is carried out on separate samples of the following four aqueous solutions using inert graphite electrodes:

1.  $\text{AgNO}_3(\text{aq})$  2.  $\text{Ca}(\text{NO}_3)_2(\text{aq})$  3.  $\text{CuSO}_4(\text{aq})$  4.  $\text{NaCl}(\text{aq})$

In which of these experiments is the metal deposited at the cathode?

- A 1 and 3 only
- B 2 and 4 only
- C 1, 2 and 3 only
- D 3 and 4 only
- E 1, 2, 3 and 4

- 20** The boiling points of the following three compounds are  $-100^{\circ}\text{C}$ ,  $-24^{\circ}\text{C}$ , and  $13^{\circ}\text{C}$  (not necessarily in that order): 1. Boron trifluoride ( $\text{BF}_3$ ) 2. Chloromethane ( $\text{CH}_3\text{Cl}$ ) 3. Boron trichloride ( $\text{BCl}_3$ )

Which option lists the compounds in order of **increasing** boiling point? (Ar values:  $\text{H} = 1$ ;  $\text{B} = 11$ ;  $\text{C} = 12$ ;  $\text{F} = 19$ ;  $\text{Cl} = 35.5$ )

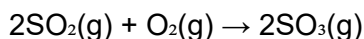
- A**  $\text{BF}_3 < \text{CH}_3\text{Cl} < \text{BCl}_3$
- B**  $\text{BF}_3 < \text{BCl}_3 < \text{CH}_3\text{Cl}$
- C**  $\text{CH}_3\text{Cl} < \text{BF}_3 < \text{BCl}_3$
- D**  $\text{BCl}_3 < \text{CH}_3\text{Cl} < \text{BF}_3$

- 21** Compound P has condensed structural formula  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ . Compound Q has condensed structural formula  $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$ .

Which statement is correct?

- A** P and Q have different molecular formulae.
- B** P and Q are structural isomers.
- C** P is an alkane but Q is an alkene.
- D** P and Q differ only in the positions of their hydrogen atoms in space.
- E** P and Q must have identical physical properties.

- 22 The reaction energy profile for the oxidation of sulfur dioxide is shown below:

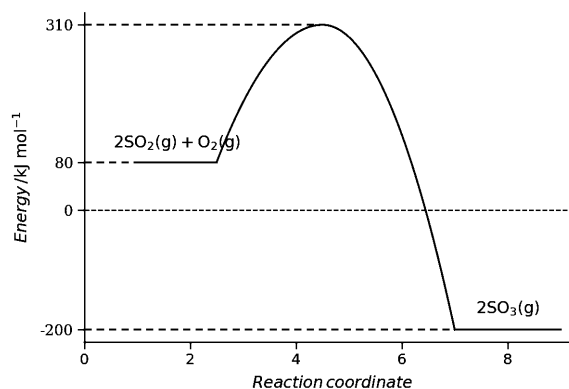


In an experiment, the thermal energy released by the complete oxidation of a sample of  $\text{SO}_2(\text{g})$  is transferred to 600 g of water. The heat transfer process is 75% efficient.

The temperature of the water increases by  $25^\circ\text{C}$ .

What mass of  $\text{SO}_2$  was reacted?

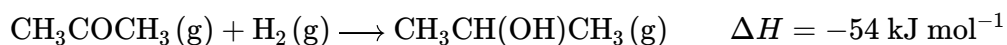
( $A_r$  values: S = 32.0, O = 16.0. Specific heat capacity of water =  $4.2 \text{ J g}^{-1}\text{C}^{-1}$ )



- A 10.8 g  
B 19.2 g  
C 21.6 g  
D 38.4 g  
E 48.0 g  
F 53.8 g
- 23 A dry mixture contains sand and copper(II) sulfate crystals. Which sequence would allow a student to obtain both the sand and pure copper(II) sulfate crystals?
- A Add water, filter, then crystallise the filtrate.  
B Heat strongly, then use a separating funnel.  
C Add water, evaporate the whole mixture to dryness, then filter.  
D Use fractional distillation, then centrifuge the residue.  
E Filter the dry mixture, then add water to the residue.

**24** The structure of propan-2-ol is shown below:

Propanone reacts with hydrogen gas to form propan-2-ol according to the following equation:



Using the mean bond enthalpies provided below, what is the mean bond enthalpy of the C=O bond in propanone?

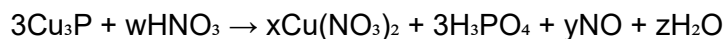
Mean bond enthalpies:

- C-C:  $348 \text{ kJ mol}^{-1}$
- C-H:  $413 \text{ kJ mol}^{-1}$
- C-O:  $358 \text{ kJ mol}^{-1}$
- H-H:  $436 \text{ kJ mol}^{-1}$
- O-H:  $464 \text{ kJ mol}^{-1}$



- A**  $332 \text{ kJ mol}^{-1}$
- B**  $440 \text{ kJ mol}^{-1}$
- C**  $745 \text{ kJ mol}^{-1}$
- D**  $853 \text{ kJ mol}^{-1}$
- E**  $1181 \text{ kJ mol}^{-1}$

**25** A chemical equation that represents the reaction of copper(I) phosphide with dilute nitric acid is:



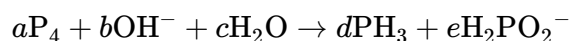
where  $w$ ,  $x$ ,  $y$ , and  $z$  are integers.

What is the value of the sum  $w + x + y + z$ ?

- A** 49
- B** 53
- C** 57
- D** 59
- E** 63

- 26** When white phosphorus,  $P_4$ , reacts with warm aqueous sodium hydroxide, it undergoes a disproportionation reaction to produce phosphine gas,  $PH_3$ , and hypophosphite ions,  $H_2PO_2^-$ .

The unbalanced ionic equation for this reaction is:



What is the simplest whole-number ratio of  $e : d$  in the balanced equation?

- A** 1 : 1  
**B** 1 : 2  
**C** 2 : 1  
**D** 1 : 3  
**E** 3 : 1  
**F** 1 : 4  
**G** 4 : 1  
**H** 3 : 4
- 27** The apparatus is used to coat a metal spoon with silver. Which of the following statements is/are correct?
1. The spoon should be connected to the negative terminal so that  $Ag^+$  ions are reduced on its surface.
  2. A silver anode can replace  $Ag^+$  ions removed from the electrolyte during plating.
  3. Alternating current is preferred because regularly reversing the electrode polarity increases the thickness of the coating.
- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3